36 Recommendations for Reporting Results in Figures, Tables, and Text

Communicating your results and ideas to the scientific community is almost as important as analyzing the data. There are many reasons why you should try to disseminate your findings in a clear and concise way. There are personal motivations—peer-reviewed publications are the currency of science and facilitate career advancement and grant funding. There are scientific motivations—showing the scientific community your results opens opportunities for constructive criticisms and collaborations and therefore will help you become a better scientist. And there are communal motivations—if you want to contribute to science and increase the corpus of human knowledge, you need to communicate your results in a manner that will disseminate your knowledge into the world and inspire others to improve their research based on your findings. Regardless of your motivations, making clear figures that are easy to interpret is a crucial part of communicating your results and ideas to the scientific community.

Probably you, like many scientists, are busy and, sadly, do not have enough time to read thoroughly all of the papers that you would like to (and probably should). Probably you have had the experience of downloading the pdf of a publication and deciding whether to read it based on the title, the abstract, and the figures. For those readers who teach or give lectures often, you may also know the experience of deciding which scientific findings to highlight in your lecture based on which publications have clear figures that can be interpreted with minimal effort.

Making excellent scientific figures is an art. I do not claim to be an artist in this respect, and I do not always make excellent or easily interpretable figures. But I have tried to think about what differentiates good figures from poor figures. This chapter contains recommendations and points of consideration for making figures. You may disagree with some of the suggestions in this chapter. Even if you agree with all of them, there is no guarantee that following them will produce excellent figures. Furthermore, there are no rules to follow to make excellent figures; some guidelines may be more or less useful depending on what message you are trying to convey in the figure. The best way to make good figures is to think carefully and critically about them. Therefore, this chapter is not a cookbook for making figures but, rather, is a list of points to consider that I hope will help you make excellent figures. Additional advice about how to show scientific results can be found in these references (Allen, Erhardt, and Calhoun 2012; Franzblau and Chung 2012), any of Tufte's books on the topic (e.g., Tufte 1983), and of course, the humorous and informative book *How to Lie with Statistics* (Huff 1991).

36.1 Recommendation 1: One Figure, One Idea

Each figure should have only one "take-home message." Expressing too many ideas within one figure makes the figure confusing and difficult to interpret. Similarly, redundantly expressing one idea over two or more figures might lead readers to wonder whether they misunderstood the idea of one or both figures. This does not mean that a figure should contain only one result—many results might be related to one idea.

If a figure has one take-home message, you should be able to write a one-sentence summary that expresses the main idea of that figure. For example, "This figure shows that stimulus-contralateral alpha suppression during an attention task is enhanced after eating a grapefruit." This figure might show several results, perhaps from several experiments and follow-up control tests, but all of the results support the one idea of that figure—eating a grapefruit boosts the attention-modulation of alpha suppression. This summary sentence could be the first sentence in the figure legend. This will help you decide whether the figure contains too much or too little information and it will also help readers quickly interpret the figure.

36.2 Recommendation 2: Show Data

One of the many advantages of EEG is that you can show data that have been only lightly processed. Showing data in figures is less common in fMRI research and in many psychology studies; instead, parameter estimates from a statistical model that reflect the fit of that model to the data are shown. Time-frequency power plots, for example, simply reflect the raw data after temporal filtering and baseline normalization. Therefore, you can and should show data in figures.

How much data you should show depends on whether the study was driven by hypothesis testing or by data exploration. If the study was strictly hypothesis driven, only the most important results need to be shown, and this might be best done with bar plots or scatterplots. For hypothesis-driven tests, showing too much data can be confusing. For example, if your study concerns lateralized visual evoked potentials, showing time-frequency plots of cross-frequency coupling at anterior electrode AF3 will only confuse readers.

On the other hand, if your study is primarily exploratory and data driven, there are a lot of results to show, and it might not be known which results will become the most relevant. Therefore, there should be many figures illustrating results using several different slices of the results hypercube (figure 3.2). The Matlab utility `tfviewerx` (available in the online Matlab code) should help you search through the results hypercube to know which slices show the results effectively.

The following is a general suggestion for how to show task-related time-frequency results. First, show one plot (e.g., a time-frequency plot and/or a topographical map) that illustrates the overall task-related effects, measured as the average activity of all conditions relative to baseline ("activity" is whatever measure is most relevant, such as power, ITPC, or connectivity). This is useful because it illustrates the electrophysiological landscape of task-related activity. Next, show plots of condition differences to highlight which of the dynamics are most relevant for the cognitive process under investigation. Use the same time-frequency-electrode characteristics that were used in the first plot to facilitate comparison. An example is shown in figure 36.1 (plate 26).

One limitation of difference plots is that it may be difficult to infer the direction of the effect for each condition. Thus, it may be useful to supplement condition difference plots with bar plots that illustrate the effect for each condition separately. An example is shown in figure 36.2. The two sets of bar plots illustrated in panel B are equally likely given the difference map in panel A, but these two patterns of results would elicit different interpretations. Note that illustrating bar plots based on condition differences is not circular inference when done appropriately: the data were selected for the bar plots because of a condition difference, and thus, the difference between the bars is biased toward being statistically significant. It would be inappropriate to perform a statistical test on the data shown in the bar plots because those data were selected in a biased manner. However, showing the bar plot to facilitate a qualitative interpretation of the condition difference is not circular inference. If you are concerned that readers might get confused about the appropriateness of how you show the data, you could mention explicitly that the data in the bar plot are selected based on their statistical significance and are thus shown to facilitate interpretation rather than to be further statistically analyzed.

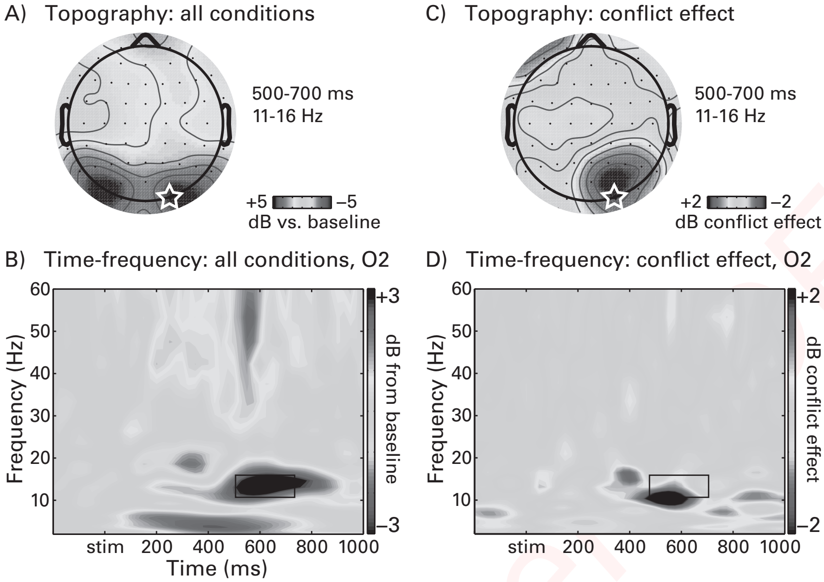

Figure 36.1 (plate 26)

Recommendation for how to show results over time, frequency, and space. Panels A and B show results from all conditions averaged together, relative to the baseline activity. These plots are useful because they show the basic task-related responses. The white star over the right posterior electrode O2 illustrates that the time-frequency plot was taken from this electrode, and the black box in the time-frequency plot illustrates the window selected for the topographical map. Panels C and D show the condition effect (in this case it was a subtraction of trials that contained response conflict from trials that contained no response conflict).

36.3 Recommendation 3: Highlight Significant Effects Instead of Removing Nonsignificant Effects

Consider figure 36.3. Panels A and B both illustrate that there is a suprathreshold effect at around the third tick on the *x*-axis. Panel A, however, reveals other features of the results that are large in magnitude and nearly statistically significant. These features of the results might be theoretically relevant or might be of interest to readers who can link those results to other literatures.

For time-frequency plots, you can show all of the data and use contour lines to illustrate statistically significant regions. An example using real data is shown in figure 36.4 (plate 27), and the online Matlab code from chapter 33 will show you how to overlay significant results

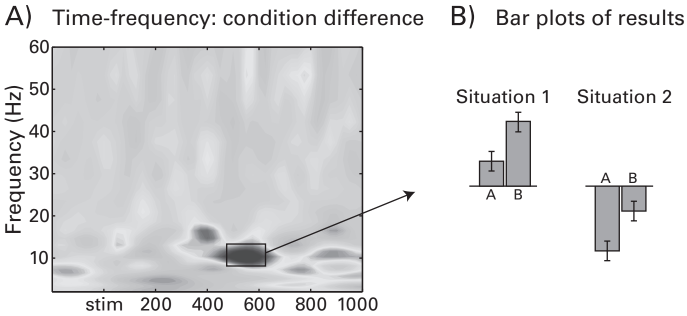

Figure 36.2

Although a condition difference plot highlights regions in time-frequency space where there are condition differences (panel A), they do not show the direction of the effect for each condition. The bar plots in panel B highlight two situations that would produce an identical condition difference map and test statistic but would have different interpretations. Because the bar plots were selected from a time-frequency region that shows a condition difference, it would be inappropriate to perform statistical tests on the data in the bar plots. However, showing the data for visual inspection is appropriate and will facilitate a deeper appreciation of the results.

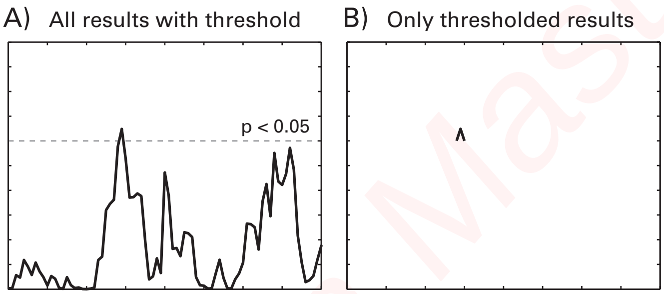

Figure 36.3

An information-rich and complex landscape (panel A) may be poorly represented by removing all subthreshold results (panel B).

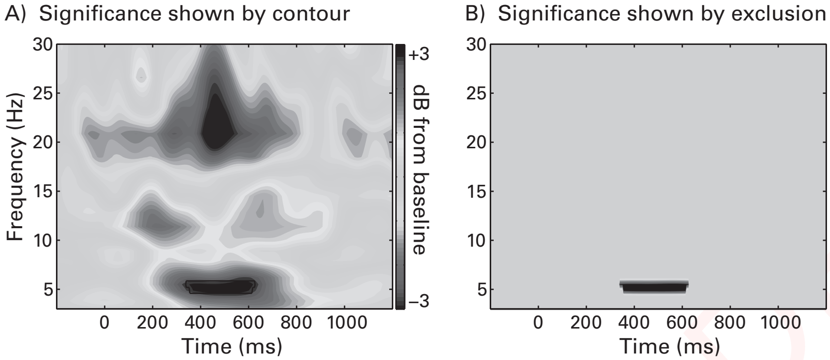

Figure 36.4 (plate 27)
Two methods of showing statistical significance in time-frequency plots. Panel A shows all results and highlights the statistically significant time-frequency regions. This is a useful representation because readers can see a broad range of dynamics and might be able to link the results to other literatures and findings. It is also useful because it shows what other effects might be present in the data that are perhaps relatively subtle, and it shows whether the effects are specific to a certain time-frequency range (see also figure 36.3). In contrast to panel A, panel B contains an alternative method of showing significant results. Here, all nonsignificant pixels are set to zero. Although this makes the significant effect more visually salient, it also obscures a considerable amount of information in the time-frequency dynamics from this electrode. Other examples of highlighting significant regions versus setting nonsignificant pixels to zero were shown in chapter 34.

using single contours. For topographical maps, you can indicate electrodes with statistically significant results by putting circles or stars on those electrodes.

The advantage of showing all results is that it allows you and readers to determine whether the statistically significant effect is an “island” or merely the tip of an iceberg. This is a nontrivial point: the interpretation of a result might be different if a significant region appears to be brief in time and localized in frequency, compared to a temporally extended and broadband effect for which one particular region happened to have relatively less variance and therefore to exceed a significance threshold.

36.4 Recommendation 4: Show Specificity (or Lack Thereof) in Frequency, Time, and Space

Imagine you are conducting an experiment about the role of motivation on conflict monitoring. Based on previous findings, you hypothesize that there will be modulations in theta-band power between the stimulus and response at electrode FCz. Technically, to test this hypothesis, you need to use only one wavelet centered at 6 Hz and analyze data only from electrode FCz.

In practice, however, it is useful to analyze activity in a range of frequencies (in this case, perhaps 2 to 40 Hz) and many electrodes. This will allow you to show time-frequency plots and topographical maps. This does not necessarily mean that you need to evaluate the statistical significance of activity in every electrode, frequency band, and time point, and it does not mean that you need to correct for multiple comparisons across all possible results that could be tested. You can still restrict your statistical analyses to the window for which you had a priori hypotheses. There are two reasons to show results from a broader time-frequency range than what your hypotheses specify.

First, it will allow you to determine whether the effect of interest is specific to the time-frequency-electrode window for which you had a hypothesis. As mentioned at the end of section 36.3, the interpretation of your finding might be different if the effect spans several frequency bands or hundreds of milliseconds rather than being isolated in one frequency band for a fairly brief period of time. Second, showing a broader range of results will allow you and readers to visually inspect a richer portion of the landscape of EEG activity. You might discover additional task-related dynamics that were not specified by your hypotheses. Post hoc data exploration should not be discouraged as long as it is labeled as such, evaluated with appropriate statistical corrections, and interpreted cautiously. Post hoc analyses and qualitative data inspection may help contextualize the findings and may provide novel links between your results and other literatures. This is particularly important when the time-frequency characteristics of the cognitive process under investigation are relatively unknown or understudied.

36.5 Recommendation 5: Use Color

Color provides an additional dimension for encoding information in figures and should be used when possible. The advantages of color for interpreting time-frequency plots can be seen in many figures in this book by comparing the grayscale figures in the main text with the color figures in the pages in the center of the book.

Choose your colors carefully. Some color combinations may look good on your computer screen but may be difficult to differentiate on someone else's computer screen or when printed on paper. The best colors to use in figures are those that are very different from each other. For example, you should use green and purple rather than sky blue and electric blue. If you are unsure whether your color combination is good, print out the figure rather than looking at it on your computer screen. It is also a good idea to print the figure in grayscale because your figures should ideally be readable to people who use black-and-white printers. In some cases color figures will not be interpretable when converted to grayscale because, for example, deep blue and hot red colors are both converted to black.

Try to make your figures color-coordinated. If you show activity for conditions A and B using the colors green and black, use those same colors for the same conditions in all figures. This will help readers compare results across analyses.

If you cannot use colors, try to use different shades of gray and change the color thickness and type (solid, dashed, or dotted). If you need to plot many lines on the same grayscale figure, use different shades of gray, such as 100%, 66%, and 33% black saturation rather than 90%, 80%, and 70%. The same advice holds for line thickness: use 0.5- and 2.0-point lines rather than 0.75- and 1.0-point lines (figure 36.5).

Most people use the color map "jet" for their analyses. This is the color map that goes from blue for large negative values to red for large positive values and green for zero values. Because this color map is so widely used and easily interpreted, you should use it as well, particularly for results that can take positive or negative values. If you are plotting a variable that cannot be negative, such as ITPC or nontransformed Granger results, you should avoid using "jet" or any other bipolar color scale. The reason is that most people expect blue colors in figures to correspond to something negative, and thus, seeing ITPC

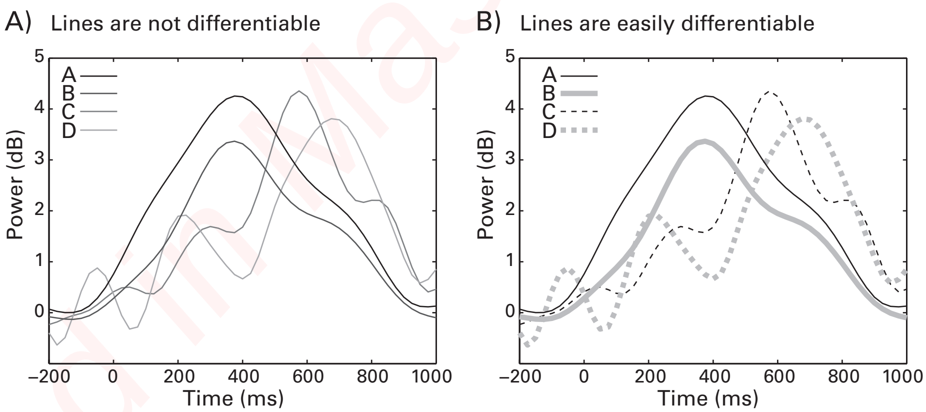

Figure 36.5
Poor use of grayscale and line styles makes it difficult to identify which time courses belong to which conditions. In panel A, grayscale saturation levels of 100%, 80%, 60%, and 40% were used for conditions A, B, C, and D. Panel B illustrates a better method for differentiating lines. In this case the four conditions are differentiated based on line thickness, solid versus dashed lines, and grayscale saturation. Even readers with poor-quality printers will see which lines correspond to which conditions.

corresponding to "negative colors" can be confusing (it would make sense, for example, in a figure with ITPC condition differences). There are several unipolar color scales, including "hot," "bone," and "grayscale." You can also define your own color scales in Matlab.

Some journals will publish color figures for free in the online version and use grayscale for the print version. Very few people walk to the university library to look at physically printed and bound journal articles, so having grayscale figures in the print version of a publication should not be a major drawback.

36.6 Recommendation 6: Use Informative Figure Labels and Captions

Some figures are surprisingly difficult to interpret. To be sure, the majority of figures are at least reasonably well documented, but it is also not rare to see a publication with figures for which you cannot determine from which electrode activity is being plotted, or what conditions, time points, or frequencies.

Ideally, the information presented in the figure should be self-evident from the figure, legend, and captions. This is particularly the case if you want to attract potential readers to your paper. If the figure legend simply reads "Figure 3. See text," many people will not follow that terse instruction but will doom the paper to sit forever in the "to-read" pile (everyone has one of those). The first line of the figure legend should be a succinct description of the main message of that figure (as discussed in section 36.1), and ideally, this should be understandable even to people who have not read the paper. For example: "Figure 3. Connectivity degree, a measure of large-scale network connectivity, increased over parietal areas after food-deprived subjects were shown pictures of grapefruits." If you use abbreviations in the figures, define them in the legend even if they are also defined in the main text, unless they are commonly used abbreviations such as ms, Hz, cm, or dB.

Label axes with meaningful information, including the scale of the values [e.g., dB, percentage change, time (ms), frequencies (Hz)]. Show a color bar and label both ends of the color bar with numbers. "Min" and "max" colorbar labels are self-evident and meaningless and should not be used. If the units are arbitrary, write "arbitrary units" or "arb. units." Figure 36.6 (plate 28) illustrates how useful informative labeling can be.

Figures published in papers are almost always smaller than you think they will be while you are making the figures on your computer. Therefore, you should use a large font size. The text should be as large as it reasonably can be without losing spatial selectivity or becoming bigger than the images. If the font size seems a bit small, it is definitely too small; if the font size seems right, it's too small; and if the font size seems too big, it's just right (see figure 36.7). Also consider that if someone else is thinking about using your figure in their talk or

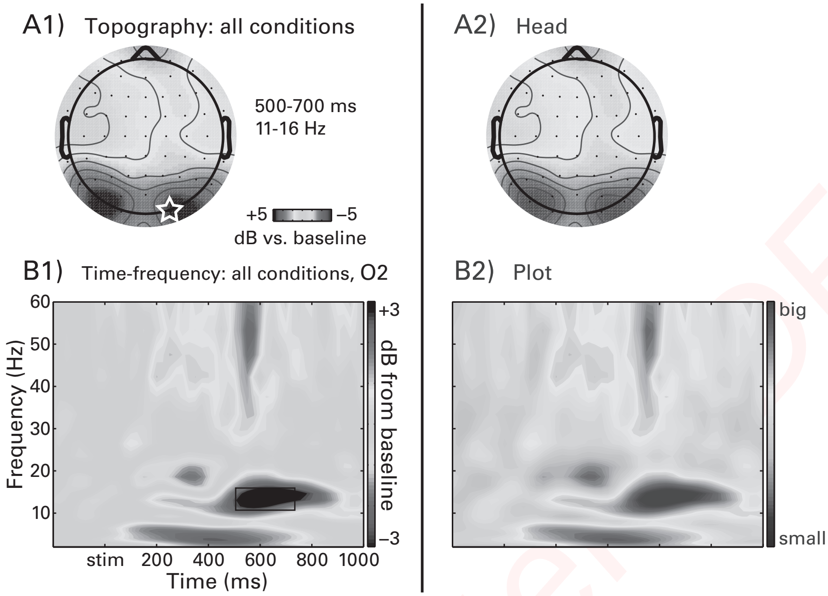

Figure 36.6 (plate 28)
Time-frequency and topographical plots that have informative (panels A1 and B1) and uninformative (panels A2 and B2) captions accompanying the plots. It is difficult or impossible to extract information from the plots in panels A2 and B2.

lecture, your figure will be pixilated and expanded to many times its original size. If the font is barely visible in the paper, it will render simply as a gray box in someone's PowerPoint slide show.

36.7 Recommendation 7: Avoid Showing "Representative" Data

Sometimes you see a figure showing what the authors claim to be representative data—a small amount of data, typically one or a few trials from one subject, one electrode, or one neuron.

There are two reasons why you should not show "representative data" unless you are illustrating how an analysis works. First, critical readers will be suspicious of representative data no matter what those data show. If the representative data look similar to the group result, readers will suspect that you hand-picked that example. If the representative data look different from the group result, readers will wonder why you picked that example, and they may question the robustness and validity of the group result. Second, the reason you collect

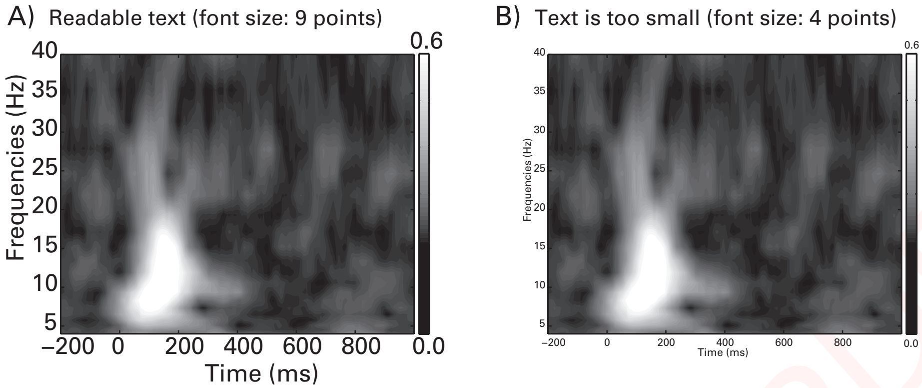

Figure 36.7
Text inside figures can easily become too small to be legible. When zoomed in to 800% on my 27-inch desktop monitor, the text labels were easily readable in both panels A and B. However, you should not assume that your readers will have the same comfortable viewing environment. When this figure is printed the text may become unreadable, and when a screen shot of this figure is taken and projected on a wall, the text in panel B will likely seem like small gray dots. Some journals make explicit recommendations for minimum font sizes, but in general, avoid using font sizes smaller than 8 points when possible.

data from many trials and from many subjects is that single trials and single subjects are not representative of the population response. In other words, critical readers will be cautious about making any inferences based on the results in a figure that shows representative data.

This does not mean that you should avoid showing single-subject data. Showing single-subject data can make a compelling case for the robustness of an effect, or it can highlight the individual variability in the results. If there are few subjects it is arguably better to show single-subject data than group-averaged results. For example, in intracranial EEG studies, each of five patients in a dataset might have electrodes placed in different brain regions. In this case you can and should show data from each subject. Furthermore, if you want to illustrate that you have clean data and can observe the effect in single subjects, show the single-subject results from all subjects (in a supplemental materials section if necessary). The argument here is that you should not show data from one subject when you collected data from 20 subjects.

The exception to this recommendation—the situation in which you should show "representative" data—is in a figure in which you are illustrating how a method works (which was done throughout this book, for example) or showing the temporal structure of the experiment. If this is the goal, do not pick data randomly; hand-pick a nice-looking piece of data that you think clearly illustrates the point. Make sure you state clearly in the figure legend that those data were selected for illustration purposes. That way, readers know that they should not interpret the result in the figure but, rather, understand the method that those data highlight. For example, data were shown in the figures in this book to illustrate how different methods work; you should not draw firm conclusions about fundamental brain mechanisms based on the results here because all of the figures are based on one subject (and in many cases, one trial).

36.8 A Checklist for Making Figures

Following is a list of questions that you can ask yourself to make sure your figures will be interpretable. Perhaps you have additional questions to add to this list, or perhaps some of these questions are not relevant for your figures. However, asking yourself these questions will help you make clear and readable figures.

- What is the main message of the figure?
- Does the figure convey one idea, several related ideas, or several different ideas?
- Can you easily discern which lines/bars/plots correspond to which conditions/groups/analyses?
- Can you easily distinguish the significant results from the nonsignificant results?
- Is the color helpful or annoying?
- Will this be readable when printed on a grayscale printer?
- Can the figure be interpreted on its own, or do you need to read the text to understand it?

36.9 Tables

Tables of activation are common in fMRI research. The tables typically list spatial *XYZ* coordinates for peak activations, a measure of effect size or statistical significance such as a *t*-value or a *p*-value, and a description of the effect that revealed the activations (e.g., condition A > condition B). Tables in fMRI papers are so common that automated programs can search journal websites and read data directly from tables of activation to use for meta-analyses (Yarkoni et al. 2011).

Tables of activation are uncommon in cognitive electrophysiology, particularly for time-frequency results. This likely reflects that most cognitive electrophysiology research is done in one of two ways: the research is hypothesis driven, and only the analyses that were specified a priori are tested and reported in the Results section; or the research is data driven and exploratory, and suprathreshold results are shown in figures and discussed in terms of their qualitative patterns. Either way, researchers rarely organize the information into easily readable tables.

The lack of tables is understandable—time-frequency-based analyses create a large search space, and it is impractical to search through the entire space for significant results to put in a table. For example, multiple-comparisons correction is only sometimes done over time points, frequencies, and electrodes simultaneously; more often, one or a few electrodes are selected, and then pixels in the time-frequency map are tested using multiple-comparisons correction, or one time window is selected and all electrodes are tested with correction for multiple comparisons.

However, tables of activation are useful for organizing results and will allow readers to inspect other findings that may have been statistically significant but not focused on in the paper. Furthermore, tables of activation will facilitate future meta-analyses. Tables should be included in cognitive electrophysiology publications. At a minimum, the following columns should be included:

1. The type of analysis (ERP, time-frequency power, ITPC, phase-amplitude coupling, etc.)
2. The statistical contrast (all conditions vs. baseline, conditions A–B, adolescents–adults, etc.)
3. The electrode label or *XYZ* coordinate (can be the center electrode if several electrodes were pooled)
4. The time point at which the effect was maximal or the time window used in the analysis
5. The frequency band at which the effect was maximal or the frequency band used in the analysis
6. A measure of effect size (dB, percentage change, *F*/*t*/*z*-statistic, etc.)

36.10 Reporting Results in the Results Section

Because time-frequency-based results can be complicated, and because it is common to perform many analyses (e.g., power, phase, connectivity), try to write the Results section as if you are guiding the reader along a path through the landscape of the results. This is particularly important if your study concerns a topic for which time-frequency-based analyses are infrequently applied, and thus readers may be unfamiliar with interpreting time-frequency-based results.

This is no trivial matter. On the one hand, you want to be open and forthcoming about the results without hiding data, or seeming to readers as if you are hiding data. On the other hand, there are simply too many results to show everything without drowning the reader in a confusing ocean of numbers, colors, and contour lines. Having clear and specific hypotheses will help you decide which results should be shown and discussed, and they will help you justify your decisions of showing some regions of the data hypercube and not other regions.

Organize the presentation of the results in a sensible way. The results could be organized according to hypotheses, or according to the time period during the task (e.g., stimulus, delay, response, feedback), or according to the type of analysis (e.g., ERP, time-frequency power, phase-based connectivity). While working on the manuscript, do not be afraid to reorganize the Results section if you think a different organization scheme might be more sensible.

The organization of the Results section also depends in part on how hypothesis driven versus exploratory the study is. The Results section of hypothesis-driven research is likely to contain statistical values, degrees of freedom, *p*-values, and statements concerning main effects and interactions. The Results section of an exploratory study, in contrast, will contain few statistical values and instead contain qualitative descriptions of the patterns of the results that exceeded a statistical significance threshold.

Related to the organization and as mentioned above in section 36.2, it is informative for readers if you describe the overall characteristics of the results first, before moving on to the condition-specific effects and hypothesis-driven analyses. These descriptions can be qualitative and need not be accompanied by statistical tests unless that is relevant. For example: “When we examined the main task effect (all conditions averaged together compared to baseline), we observed increases in theta-band power over midfrontal electrodes from around 200–800 ms post-stimulus-onset, alpha-band suppression over occipitoparietal areas around 500–2000 ms, and beta-band suppression over motor areas around 1000–1700 ms.” These general descriptions will help readers form a framework of cortical dynamics within which to place the condition-specific results.

Introduce each set of analyses to help the reader understand what results they are about to read. This is particularly important for papers with many complex sets of analyses. For example, you can start each paragraph or subsection of the Results with “the purpose of the next set of analyses was to examine whether…”

Relatedly, having a closing sentence will also help readers consolidate what they just read. For example, “Taken together, these statistical tests generally confirm our hypothesis that food-deprived subjects exhibit enhanced local connectivity when viewing pictures of grapefruits, although, in contrast to our expectations, the results were observed primarily over parietal areas instead of occipital areas.”

Avoid interpreting the results or speculating about the meaning of the results when reporting condition comparisons and statistical effects. This is the purpose of the Discussion section; the Results section should ideally be constrained to an objective description of the significant and, when relevant, nonsignificant results. In some cases it is necessary to interpret results, point out that a (perhaps unexpected) finding is similar to a finding published previously, or motivate a post hoc analysis that is based on an interpretation of a result. But statements about interpretations of what the results might mean should be used sparingly in the Results section.

A final matter discussed here with regard to the Results section is reporting *p*-values. You might think this is an easy issue: report *p*-values associated with statistical tests. For hypothesis testing, this is good advice: you should report the *p*-value associated with each statistical test. Furthermore, you should report the *p*-value itself rather than simply noting whether the *p*-value was less than 0.05. This is because, for example, the *p*-values 0.048 and 0.0000048 are both statistically significant using a threshold of 0.05 but might be interpreted differently with respect to how strongly the results support the hypotheses.

On the other hand, if your study involves exploratory data analyses, there may be no theoretical framework within which to compare qualitatively findings associated with *p*-values of 0.048 and 0.0000048. Therefore, you could report no *p*-values at all but instead show the results in figures such that suprathreshold pixels or electrodes are outlined or otherwise demarcated and then describe qualitatively the patterns of significant results. Indeed, if mapwise results are thresholded at a specific *p*-value, reporting *p*-values associated with the results would be biased because the pixels would be selected based on being statistically significant. In the case of selecting data based on a significant result, it would be impossible to report a *p*-value that you do not already know exceeds the significance threshold, and therefore, reporting *p*-values might be circular inference.